HW2 - Math 548/448

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1 3.1 Bisection Method

5. Give an example in which $a_0 = a_1 < a_2 = a_3 < a_4 = a_5 < a_6 = ...$

One possible example is the sequence $1 = 1 < 2 = 2 < 3 = 3 < 4 = ...$ where $a_0 = 1, a_1 = 1, a_2 = 2, a_3 = 2, a_4 = 3, a_5 = 3, a_4 = 4...$
and certainly $a_1 = 1 < a_2 = 2 = a_3 = 2 < a_4 = 3 = a_5 = 3 < a_6 = 4...$

Explicit Formula (Using the floor function): $a_n = \lfloor n/2 \rfloor + 1$

13. If the bisection method is used (in single precision) on the Marc-32 starting with the interval [128,129], can we compute the root with absolute precision $<$ 0.000001? Answer the same question for relative precision.

In normalized base-2 scientific notation, the boundaries of the search interval are exactly $128 = 2^7$ and $129 = (2^7 + 2^0)$.
For any x $\in$ [128,129], $x = (1, f_1, f_2, ...f_2 3)_2 \times 2^7$

ULP (Unit in the last place) $= 2^{-23} \times 2^7 = 2^{-16}$

During the bisection method, the working interval $[a, b]$ is repeatedly halved. However, the physical computation stagnates when a and b become adjacent machine numbers. At this limit, evaluating the midpoint $c = a + (b - a)/2$ results in no new information, as floating-point rounding forces c to snap back to either a or b. Consequently, the smallest non-zero interval width the machine can resolve around a root in this range is exactly 2^{-16}.Assuming the true root x^* is not an exact floating-point number, the tightest bound we can guarantee is to narrow it down to this final interval. Returning the midpoint of this interval yields a maximum absolute error of half the interval width:

Max Absolute Error: $\frac{1}{2} \times 2^{-16} = 2^{-17}$

As a decimal this equates too $\approx$ 0.000007629 which is greater than our tolerance of 0.000001.

Therefore, the machine cannot physically represent the boundaries tight enough to guarantee the target absolute precision.

Relative precision normalizes the absolute error against the magnitude of the true root. Since the root lies within the interval [128, 129], its absolute minimum magnitude is bounded by $x^* \geq 128 = 2^7$. Using the maximum absolute error established above, the maximum relative error is bounded by:
Max Relative Error $= \frac{MaxAbsoluteError}{|x^*|} \leq \frac{2^{-17}}{2^7} = 2^{-24}$

Converting this bound to decimal form:

$$2^{-24} = \frac{1}{16777216} \approx 0.0000000596$$

Because 0.0000000596 is strictly less than 0.000001, the guaranteed relative precision falls well within the required threshold.Therefore, using the bisection method on the Marc-32 starting with the interval [128, 129], you cannot compute the root with an absolute precision of < 0.000001, but you can compute it with a relative precision of < 0.000001.

3. Computer problem: Find the root of $f(x) = x - \tan(x)$ on [1,2] using the bisection algorithm.

The code returns NaN (Not a Number) no matter the number of iterations or tolerance as expected since no roots exist on the interval [1,2] for this particular function.

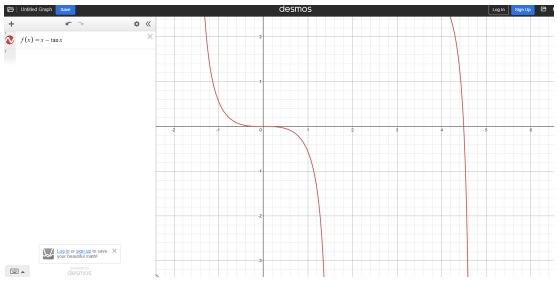


Figure 1: Graph of f(x) = x-tan(x)

```python
import warnings
import math

def f(x):  # 1 usage
    return x - math.tan(x)

def bisection_method(f, a, b, tol, max_iterations):  # 1 usage
    # Check if the function has the same sign at the endpoints
    # f(a) * f(b) > 0 is the standard Python equivalent to checking if signs match
    if f(a) * f(b) > 0:
        raise ValueError("Function has the same sign at the endpoints. No root in this interval.")

    # Initialize the variables
    iterations = 0
    root = math.nan

    # Perform bisection iterations
    while iterations < max_iterations:
        # Compute the midpoint of the interval
        c = (a + b) / 2.0

        # Check if the root has been found within the tolerance
        if abs(f(c)) < tol:
            root = c
            return root, iterations

        # Update the interval by narrowing it down
        if f(c) * f(a) > 0:
            a = c
        else:
            b = c

        iterations += 1  # Increment the iteration counter

    # Maximum number of iterations reached without finding the root
    warnings.warn(
        "Maximum number of iterations reached. The root may not have been found within the desired tolerance.")

    return root, iterations

print(bisection_method(f, a: 1, b: 2, tol: 0.00001, max_iterations: 100))
```

Figure 2: Bisection Algorithm Implementation

2 3.2 Newton's Method

3. Prove that if Newton's Method is used on a function f for which f'' is continuous and f(r)= $0 \neq f'(r)$, then the $\lim_{n\to\infty} e_{n+1}e_n^{-2}$ exists and equals $\frac{f''(r)}{[2f'(r)]}$ How can this fact be used in a program to test whether convergence is quadratic?

Let r be a root of f such that $f(r) = 0$ and $f'(r) \neq 0$. Let $e_n = x_n - r$ denote the error at the n-th iteration of Newton's Method. The iteration is defined as:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} e_{n+1} = e_n - \frac{f(x_n)}{f'(x_n)} \tag{1}$$

Expanding $f(r)$ around x_n using Taylor's Theorem yields:

$$0 = f(r) = f(x_n) + f'(x_n)(r - x_n) + \frac{1}{2}f''(\xi_n)(r - x_n)^2 \tag{2}$$

where ξ_n is strictly bounded between x_n and r. Substituting $r - x_n = -e_n$:

$$0 = f(x_n) - f'(x_n)e_n + \frac{1}{2}f''(\xi_n)e_n^2 \tag{3}$$

Dividing by $f'(x_n) \neq 0$ and noting from (1) that $\frac{f(x_n)}{f'(x_n)} = e_n - e_{n+1}$:

$$0 = (e_n - e_{n+1}) - e_n + \frac{f''(\xi_n)}{2f'(x_n)}e_n^2 \tag{4}$$

Canceling e_n and rearranging gives:

$$e_{n+1} = \frac{f''(\xi_n)}{2f'(x_n)}e_n^2 e_{n+1}e_n^{-2} = \frac{f''(\xi_n)}{2f'(x_n)} \tag{5}$$

Assuming convergence ($x_n \to r$), the Squeeze Theorem dictates $\xi_n \to r$. Given f' and f'' are continuous and $f'(r) \neq 0$, taking the limit as $n \to \infty$ gives:

$$\lim_{n\to\infty} e_{n+1}e_n^{-2} = \frac{f''(r)}{2f'(r)} \tag{6}$$

Because the exact root r (and therefore e_n) is unknown, we approximate the error using successive differences: $e_n \approx \Delta x_n = x_{n+1} - x_n$. Quadratic convergence is tested algorithmically.

Evaluate the ratio of successive step magnitudes:

$$C_n = \frac{|x_{n+1} - x_n|}{|x_n - x_{n-1}|^2} \tag{7}$$

For quadratic convergence, the sequence C_n will stabilize to a non-zero constant $C = \left|\frac{f''(r)}{2f'(r)}\right|$.

We can also dynamically compute the order of convergence p_n using logarithmic ratios of successive steps:

$$p_n = \frac{\ln(|x_{n+1} - x_n|/|x_n - x_{n-1}|)}{\ln(|x_n - x_{n-1}|/|x_{n-1} - x_{n-2}|)} \tag{8}$$

For quadratic convergence, $\lim_{n\to\infty} p_n = 2$.

4. (Steffenson's method) Consider the iteration formula: $x_{n+1} = x_n - f(x_n)/g(x_n)$ where

$$g(x) = [f(x + f(x)) - f(x)]/f(x) \tag{9}$$

Show that this is quadratically convergent under suitable hypotheses.

Let r be the simple root ($f(r) = 0, f'(r) \neq 0$) and $e_n = x_n - r$. Denoting $f' = f'(r)$ and $f'' = f''(r)$, we expand via Taylor series:

$$f(x_n) = f'e_n + \frac{1}{2}f''e_n^2 + \mathcal{O}(e_n^3)$$

$$f(x_n+f(x_n)) = f(r+e_n+f(x_n)) = f'(1+f')e_n+\frac{1}{2}f''(1+f')^2e_n^2+\frac{1}{2}f'f''e_n^2+\mathcal{O}(e_n^3)$$

Subtracting $f(x_n)$ and dividing by $f(x_n)$ yields the denominator $g(x_n)$:

$$g(x_n) = \frac{(f')^2e_n + \frac{1}{2}f'f''(3 + f')e_n^2 + \mathcal{O}(e_n^3)}{f'e_n + \frac{1}{2}f''e_n^2 + \mathcal{O}(e_n^3)}$$

$$g(x_n) = f' + f''\left(1 + \frac{1}{2}f'\right)e_n + \mathcal{O}(e_n^2)$$

Substitute into $e_{n+1} = e_n - \frac{f(x_n)}{g(x_n)}$ and apply the geometric expansion $(1+z)^{-1} \approx 1 - z$:

$$e_{n+1} = e_n - \frac{f'e_n + \frac{1}{2}f''e_n^2}{f' + f''\left(1 + \frac{1}{2}f'\right)e_n} + \mathcal{O}(e_n^3)$$

$$e_{n+1} = e_n - e_n\left(1 + \frac{f''}{2f'}e_n\right)\left[1 - \frac{f''}{f'}\left(1 + \frac{1}{2}f'\right)e_n\right] + \mathcal{O}(e_n^3)$$

$$e_{n+1} = \frac{f''}{2f'}(1 + f')e_n^2 + \mathcal{O}(e_n^3)$$

Because e_{n+1} is proportional to e_n^2, the sequence converges quadratically.

7. Define $x_0 = 0$ and $x_{n+1} = x_n - [(\tan(x_n) - 1)/\sec^2(x_n)]$

What is the $\lim_{n\to\infty} x_n$ in this example? Relate this to Newton's Method.

Newton's Method formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

General formula for problem:

$$x_{n+1} = x_n - \frac{\tan(x_n) - 1}{\sec^2(x_n)}$$

We then conclude

$$f(x) = \tan(x) - 1$$

$$f'(x) = \frac{d}{dx}(\tan(x) - 1) = \sec^2(x)$$

Because Newton's Method iterates until it finds a value where $f(x) = 0$, the limit of the sequence x_n as $n \to \infty$ will be a solution to:

$$\tan(x) - 1 = 0$$

$$\tan(x) = 1$$

The general solution for this trigonometric equation is $x = \frac{\pi}{4} + k\pi$ for any integer k. To determine which specific root the sequence converges to, we evaluate the first iteration using the given initial guess $x_0 = 0$:

$$x_1 = 0 - \frac{\tan(0) - 1}{\sec^2(0)}$$

$$x_1 = 0 - \frac{-1}{1} = 1$$

From $x_1 = 1$, the sequence converges rapidly to the nearest root. Because 1 radian is close to $\frac{\pi}{4}$ (which is approximately 0.785). Therefore, $\lim_{n\to\infty} x_n = \frac{\pi}{4}$.

3 3.4 Fixed Points and Functional Iteration

1. If F is contractive from [a,b] and $x_{n+1} = F(x_n$, with $x_0 \in [a, b]$, then $|x_n - s| \leq C\lambda^n$ for an appropriate C. Prove this and give an upper bound for C. Here s is the fixed point of F.

Let F have contraction constant $\lambda \in (0, 1)$. Since $F(s) = s$ and $x_n = F(x_{n-1})$:

$$|x_n - s| = |F(x_{n-1}) - F(s)| \leq \lambda |x_{n-1} - s|$$

Iterating this inequality n times yields:

$$|x_n - s| \leq \lambda^n |x_0 - s|$$

Thus, the inequality holds for $C = |x_0 - s|$. Upper Bounds for C.Domain Bound Since $x_0, s \in [a, b]$, the maximum distance is trivially bounded by the interval length:

$$C \leq b - a$$

. A Priori Bound (Computable)Using the triangle inequality and the contraction property $|x_{k+1} - x_k| \leq \lambda^k |x_1 - x_0|$ for any $m > 0$:

$$|x_m - x_0| \leq \sum_{k=0}^{m-1} |x_{k+1} - x_k| \leq |x_1 - x_0| \sum_{k=0}^{m-1} \lambda^k$$

Taking the limit as $m \to \infty$ (where $x_m \to s$), the geometric series converges, yielding:

$$C \leq \frac{|x_1 - x_0|}{1 - \lambda} \leq \frac{b - a}{1 - \lambda}$$

2. Prove that if $F : [a, b] \to R$, if F' is continuous, and if $|F'(x)| < 1$ on [a,b], then F is a contraction. Does F necessarily have a fixed point?

To prove F is a contraction, we must find a constant $c \in [0, 1)$ such that for all $x, y \in [a, b]$, $|F(x) - F(y)| \leq c|x - y|$.Because F' is continuous on the closed and bounded interval $[a, b]$, the Extreme Value Theorem guarantees that $|F'|$ achieves a maximum value on $[a, b]$.Let $c = \max_{x \in [a,b]} |F'(x)|$.Since we are given that $|F'(x)| < 1$ for all $x \in [a, b]$, this maximum value must be strictly less than 1. Therefore, $0 \leq c < 1$.By the Mean Value Theorem, for any $x, y \in [a, b]$ with $x \neq y$, there exists some z strictly between x and y such that:

$$F(x) - F(y) = F'(z)(x - y)$$

Taking the absolute value of both sides yields:

$$|F(x) - F(y)| = |F'(z)||x - y|$$

Because $z \in [a, b]$, we know $|F'(z)| \leq c$. Substituting this inequality gives:

$$|F(x) - F(y)| \leq c|x - y|$$

This holds trivially if $x = y$ as well. Since $0 \leq c < 1$, F satisfies the formal definition of a contraction on $[a, b]$.

No, F does not necessarily have a fixed point. Counterexample:
Let $[a, b] = [0, 1]$ and define $F : [0, 1] \to R$ by:

$$F(x) = \frac{1}{2}x + 2$$

The derivative is $F'(x) = \frac{1}{2}$, which is continuous.$|F'(x)| = \frac{1}{2} < 1$ on $[0, 1]$, making F a contraction.A fixed point requires $F(x) = x$, meaning $\frac{1}{2}x + 2 = x \to x = 4$.Since $4 \notin [0, 1]$, the function is only defined for inputs where $F(x) \neq x$, meaning it has no fixed point.

4. Show that these functions are contractive on the indicated intervals. Determine the best values of λ in Equation (2).

a. $(1 + x^2)^{-1}$ on an arbitrary interval

By the Mean Value Theorem, $f(x) = (1 + x^2)^{-1}$ is contractive on an interval I if $\sup_{x \in I} |f'(x)| = \lambda < 1$. To find the best universal constant λ, we maximize $|f'(x)|$ over R.The first and second derivatives are:

$$f'(x) = \frac{-2x}{(1 + x^2)^2}, \quad f''(x) = \frac{6x^2 - 2}{(1 + x^2)^3}$$

Setting $f''(x) = 0$ yields critical points at $x = \pm\frac{1}{\sqrt{3}}$. Evaluating $|f'(x)|$ at these points gives:

$$\left| f'\left(\pm\frac{1}{\sqrt{3}}\right) \right| = \frac{2/\sqrt{3}}{(1 + 1/3)^2} = \frac{3\sqrt{3}}{8}$$

Since $\lim_{x \to \pm\infty} f'(x) = 0$, this local maximum is the global supremum.Because $\lambda = \frac{3\sqrt{3}}{8} = \sqrt{\frac{27}{64}} < 1$, $f(x)$ is contractive on R and thus on any arbitrary interval $I \subseteq R$.The best general contraction constant is $\lambda = \frac{3\sqrt{3}}{8}$.

c. $\tan^{-1} x$ on an arbitrary closed interval excluding 0

We can do something similar for the function $\tan^{-1} x$.
By the Mean Value Theorem, $f(x) = \tan^{-1} x$ is contractive on an interval I if $\sup_{x \in I} |f'(x)| = \lambda < 1$.The first derivative is:

$$f'(x) = \frac{1}{1 + x^2}$$

For all $x \neq 0$, $1 + x^2 > 1$, which strictly bounds $|f'(x)| < 1$. The maximum of $|f'(x)|$ over any domain occurs at the point closest to 0.Let I be an arbitrary closed interval excluding 0. Because I is closed, it contains a point closest to the origin. Let $c = \min_{x \in I} |x|$. Since $0 \notin I$, we know $c > 0$.The supremum of the derivative on I is achieved at this minimum distance:

$$\lambda = \max_{x \in I} \frac{1}{1 + x^2} = \frac{1}{1 + c^2}$$

Since $c > 0$, it immediately follows that $\lambda = \frac{1}{1+c^2} < 1$. Therefore, $f(x)$ is contractive on I.The best contraction constant for the interval I is $\lambda = \frac{1}{1+c^2}$, where $c = \min_{x \in I} |x|$.

19. If the method of functional iteration is applied the the function $F(x) = 2 + (x-2)^4$, starting at $x = 2.5$, what order of convergence results? Find the range of the starting values for which this functional iteration converges. Note that 2 is a fixed point.

Let $e_n = x_n - 2$ represent the error at step n for the fixed point $p = 2$. Substituting this into the iteration formula yields:

$$e_{n+1} = x_{n+1} - 2 = (2 + (x_n - 2)^4) - 2 = e_n^4$$

Because $e_{n+1} = e_n^4$, the sequence exhibits quartic convergence (order 4). (Verified by derivatives: $F'(2) = F''(2) = F'''(2) = 0$, while $F^{(4)}(2) = 24 \neq 0$.)
For the error sequence e_n to converge to 0, the initial error must be strictly less than 1:

$$|e_0| < 1 | x_0 - 2| < 11 < x_0 < 3$$

Boundaries: If $x_0 = 1$ or $x_0 = 3$, the first iteration yields $F(x_0) = 3$, and the sequence remains at the unstable fixed point 3.
Outside $[1, 3]$: $|e_0| > 1$, causing the sequence e_n^4 to diverge to infinity.The iteration converges to the fixed point 2 for $x_0 \in (1, 3)$, and converges generally (to either 2 or 3) for the closed interval $x_0 \in [1, 3]$.